New Paleomagnetic Results from the Upper Belt – Purcell Supergroup of Alberta

M. E. EVANS, D. K. BINGHAM, AND E. W. MCMURRY

Institute of Earth and Planetary Physics, University of Alberta, Edmonton, Alberta T6G 2E1

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Stable remanent directions have been obtained from fifteen sites located in the Purcell, Shepard and Kintla Formations of the upper Belt – Purcell Supergroup. The geomagnetic polarity sequence revealed by these and other studies suggests that the frequency of reversals was probably an order of magnitude lower than that observed over the last few million years. The polarity transitions have considerable potential as regional, and possibly, global, time lines for geological correlation.

The paleomagnetic pole obtained (142 °W, 17 °S, $dp = 4°$, $dm = 7°$) agrees well with other results from the Belt Supergroup. It falls in a cluster of poles derived from various geological provinces of the Canadian Shield, which may imply that the shield was an integral unit as early as 1400 m.y. ago. However, the constancy of the pole position obtained from Belt rocks thought by some workers to be as young as 1100 m.y. conflicts with the large northward polar excursion (the Logan Loop) derived from other provinces. Possible explanations of this apparent conflict are discussed.

Des directions remanentes stables ont été obtenues de quinze sites localisés dans les formations Purcell, Shepard et Kintla de la partie supérieure du supergroupe Belt – Purcell. La séquence de polarité géomagnétique révélée par ces études et des expériences antérieures suggèrent que la fréquence des inversions était probablement d'un ordre de grandeur moindre, que celle observée depuis les dernières quelques millions d'années. Les transitions de polarité ont un potentiel considérable comme lignes de temps régionale, et possiblement globale, pour les corrélations géologiques.

Le pôle paléomagnétique obtenu (142 °W, 17 °S, $dp = 40$, $dm = 7°$) concorde bien avec d'autres résultats du supergroupe Belt. Il se situe dans un groupe de pôles provenant de différentes provinces géologique du Bouclier canadien, ce qui peut impliquer que le bouclier formait déjà un seul bloc intégral vers 1400 m.a. Cependant la position régulière du pôle obtenu des roches de Belt, considérées par quelques chercheurs comme étant aussi jeune que 1100 m.a., entre en conflit avec la grande excursion polaire nordique (la boucle de Logan) obtenue d'autre provinces. Des explications possibles de ce conflit apparent sont discutées. [Traduit par le journal]

Introduction

Early paleomagnetic surveys (Collinson and Runcorn 1960; Black 1963) involving little or no laboratory tests of stability suggest that the geomagnetic pole occupied a more or less constant position throughtout the time interval represented by the Belt – Purcell Supergroup. It also appears that geomagnetic polarity reversals took place, but at a rate very much slower than that prevailing over the last few million years. The work described in this paper represents the first results of a program of sampling throughout the Belt basin with the objective of determining the detailed temporal behavior of the field. Once established, this chronology will have direct application to the determination of global tectonic events and to the history of the Earth's core, and will probably serve as an important stratigraphic tool.

The literature contains many excellent papers on the geology of the Belt basin (*e.g.*, Smith and Barnes 1966; Harrison 1972) and therefore only a brief summary is given here. The so-called "Belt basin" is really the present remains of sediments deposited in an epicratonic re-entrant of a much larger sea that extended from California to Alaska. However, the most extensive and least disturbed outcrops occur near the U.S. – Canadian border in Montana, Idaho, Alberta and British Columbia. Reported thicknesses range up to 67 000 ft (20 435 m), and the sediments consist mostly of siltstones, argillites and quartzites with subordinate carbonate rocks.

The Precambrian strata in the southern Canadian Cordillera have been traditionally referred to as the Purcell and Windermere Groups, but Smith and Barnes (1966) define the Belt Supergroup as excluding the Windermere Group

of Canada. They therefore regard the Belt and Purcell Supergroups as "precise stratigraphic equivalents" (p. 1399). Detailed formation names differ from locality to locality, but Harrison (1972) has suggested a simplified nomenclature for the entire Belt – Purcell Supergroup. In order to facilitate comparison with earlier work we have retained the older local nomenclature (*e.g.*, Price 1964) and have included Harrison's names in brackets where appropriate.

Considerable radiometric data exist for Belt sediments, and a recent review, emphasizing current problems, has been given by Obradovich and Peterman (1973). The age of the formations sampled in this work assumes some importance when the geophysical implications of the paleomagnetic results are considered. The relevant details are therefore discussed further below. At this point, it is sufficient to state that the entire Belt Supergroup probably represents a time interval as long as the Phanerozoic, being bracketed by ages of about 1700 m.y. and 800 m.y. The Missoula Group comprises the upper part of the Belt Supergroup, and the formations studied here (Purcell Lava, Shepard and Kintla Formations) are correlated with the lower part of the Missoula Group. According to this scheme, the results reported here should be associated with the 1100 m.y. isochron obtained by Obradovich and Peterman (1968). However, as these authors have pointed out, "major problems remain in defining the geochronology of the Belt" (1973, p. 9), and in the discussion at the end of this paper we point out how the paleomagnetic results may assist in the solution of some of these problems.

Methods

A total of 48 separately oriented samples were collected in southwestern Alberta from 19 stratigraphically distinct sites. Three sites are located in the Purcell Lava, eight in the Shepard Formation and eight in the Kintla (Mount Shields) Formation. A summary of the stratigraphic relationships appears later in the paper (Fig. 2). The upper seven of the Shepard and all of the Kintla sites occur along two ridges about 1 km southwest of Mt. Gladstone (114°14′W, 49°18.5′N). The remaining sites are situated 1.5 km northeast of North Kootenay Pass (114°33.8′W, 49°24′N). These two localities correspond roughly to the locations of Black's (1963) samples DGN19 through DGN31 and of samples DN1 through DN6 respectively.

Three samples were taken from each of the Shepard and Purcell sites (only 2 were taken from one Purcell site), but frost action leading to separation along shaly partings made collection of suitable blocks very difficult and caused this number to be reduced to 2 samples at each Kintla site. Samples were oriented by means of solar bearings and/or by geographical sight points tied to solar bearings.

The uneven stratigraphic distribution of the sampling sites (see Fig. 2) is a consequence of the limited occurrence of red horizons and of our decision to concentrate sampling of those strata for which no earlier paleomagnetic data exist, namely the Shepard Formation and member B (Hage 1943; Price 1964) of the Kintla Formation.

The complete study involves about 750 measurements of remanence which were made on a PAR model SM-1 spinner magnetometer. Routine partial demagnetization was carried out on all specimens used in the final analysis. Alternating field (a.f.) demagnetization was performed with an apparatus similar to that described by McElhinny (1966). Thermal cleaning was accomplished using a small oven in a room-sized magnetic shield (Patton and Fitch 1962; Reid 1972); the magnetic field over the volume occupied by the specimens during thermal cleaning studies is less than 5 γ, and the temperature is known to $\pm$ 5°C.

Results

Results from the three formations studied are reported separately, and discussion is postponed until later in the paper.

(a) Purcell Lava

This formation consists of a series of chloritized andesite flows ranging up to 150 m in thickness. Except for a basal unit of pillowed andesite, well defined flows are seldom distinguishable.

Samples were collected at three sites; one specimen from each sample was treated thermally and one magnetically. No useful distinction can be made between the effects of the two types of treatment on specimens from site GCA, which show very little angular change between NRM and "end-points". The tightest grouping occurs after treatments of 450°C and 400 Oe, but

TABLE 1. Site mean paleomagnetic results after correction for geological dip

Site (strike, dip)	N	S	J	T	D	I	k	R
Kintla Formation								
GBH(113 32)	2	4	40–44	600	222.9	+43.3	981.8	1.9990
GBG(091 33)	2	4	13–23	600	210.3	+26.4	192.9	1.9948
GBF(097 32)	1	2	1.6–1.7	400	(083.2	+08.6	—	—)
GBE(087 31)	2	3	39–46	600	206.2	+41.1	19.9	1.9497
GBD(087 39)	2	4	0.29–0.72	400	(221.2	+18.9	3.2	1.6908)
GBC(093 38)	2	4	62–93	600	210.3	+31.3	74.0	1.9865
GBA(099 34)	2	4	15–22	600	214.4	+35.3	67.1	1.9851
GBB(099 36)	2	4	39–57	600	025.9	−49.6	290.5	1.9966
Shepard Formation								
GAG(104 40)	3	6	2.0–5.3	600	019.3	−39.1	36.8	2.9456
GAF(095 34)	3	6	1.1–2.8	600	009.8	−34.8	15.2	2.8682
GAE(088 35)	3	6	4.0–14	600	030.3	−23.4	23.4	2.9145
GAD(093 37)	3	6	0.70–20	600	012.1	−31.6	48.6	2.9589
GAC(080 37)	3	6	38–50	600	024.6	−27.0	105.2	2.9810
GAB(092 35)	3	6	7.0–13	600	041.1	−34.7	60.4	2.9669
GAA(087 33)	3	6	1.5–5.8	600	011.0	−43.4	10.8	2.8145
GCD(167 30)	3	5	2.0–3.2	600	055.4	−29.1	36.5	2.9452
Purcell Lava								
GCC(167 30)	3	3	14–30	600	031.2	−20.6	183.1	2.9891
GCB(167 30)	3	3	250–260	600	030.9	−24.6	697.6	2.9971
GCA(167 30)	2	2	26–30	450	023.4	−21.9	1896.2	1.9995

NOTES: N = number of samples; S = number of specimens; J = intensity range of NRM (10^{-6} emu cm^{-3}); T = temperature employed (°C); D, I = Declination, Inclination; $k = N\text{-}1/N\text{-}R$; R = length of resultant of N unit vectors.
Data in brackets are rejected in final analysis (see text). Parenthesized numbers following the site names represent the present attitude of the strata measured at each site. Strike is given East of true North and dip is measured clockwise from the horizontal when facing along strike.

precisions are high throughout and the site mean is always within 10° of the direction given in Table 1. It is noteworthy that as much as 64% of the remanence has blocking temperatures exceeding 600°C, presumably due to the presence of hematite.

Specimens from site GCB possess poorly grouped NRM vectors (k = 8.6), but considerable improvement takes place after magnetic and thermal cleaning. Highest precisions occur after 600°C and 800 Oe, and again hematite is indicated as a significant, if not dominant, remanence-carrier since up to 60% of the NRM remains after 600°C.

Behavior of specimens from site GCC is complex. Sample GCC03 exhibits stable behavior throughout laboratory treatments up to 1800 Oe and 660°C. The other two samples are unstable beyond 400 Oe, but during thermal treatment individual specimens exhibit two polarities, showing progressively reversed, then normal, then reversed, directions. This behavior could be due to field reversals, self-reversal or chemical change during laboratory heating, but the present data do not permit an unequivocal solution. Directions are best grouped at 660°C and 300°C. Again, up to 63% of the remanence remains after heating to 600°C.

The significance of hematite as a remanence-carrier in the Purcell Lava lies in the fact that it could imply substantial remagnetization. However, evidence from textural studies of thin sections suggests that the hematite formed during the initial cooling of the lavas (R. A. Burwash, pers. comm. 1974).

At the locality sampled the Purcell Lava is approximately 50m thick and the stratigraphic intervals between the three sites are both about 20 m. Site GCC is situated in highly vesicular material and is probably within 5 m of the upper surface of the lava. There is no evidence that more than one flow is represented at the exposure sampled. Even if the entire thickness represents a single flow it is unlikely that cooling to below typical Curie temperatures lasted more than a few tens of years (Jaeger 1957). Therefore, assigning unit weight to each site would overemphasize the particular time interval involved. Thus, unit weight is given to the whole lava (Fig. 1). This is done by averaging the three site means rather than by averaging individual sample directions since the latter procedure

TABLE 2. Statistical results for various groups of site means

	B	D	I	k	R	α_{95}	Treatment	Comments
Kintla Formation								
1.	6	211.7	+38.0	69.5	5.9281	8.1	600°C	GBB reversed; GBD, GBF omitted
2.	6	211.6	+35.7	61.4	5.9185	8.6	800 Oe	One specimen per site
Shepard Formation								
3.	8	025.8	−33.8	29.4	7.7618	10.4	600°C	
4.	8	032.0	−30.8	26.3	7.7343	11.0	800 Oe	One specimen per site
Purcell Lava								
5.	3	028.5	−22.4	314.3	2.9936	7.0	th.	GCA(450°), GCB(600°), GCC(660°)
6.	3	026.1	−26.6	38.7	2.9483	20.1	a.f.	GCA(400Oe), GCB(800Oe), GCC(300Oe)
Polarity Groups								
7.	5	212.6	+35.6	90.6	4.9559	8.1	th.	All normal sites
8.	10	026.1	−34.3	30.4	9.7038	8.9	th.	All reversed sites
Grand Mean								
9.	15	045.1	−63.6	33.1	14.5772	6.7	th.	Before tectonic correction
10.	15	028.3	−34.7	39.4	14.6446	6.2	th.	Best overall summary (see text)

NOTES: B = number of sites; R = length of resultant of B unit vectors; α_{95} = half-angle of cone of 95% confidence (Fisher 1953); D, I, k are as in Table 1.

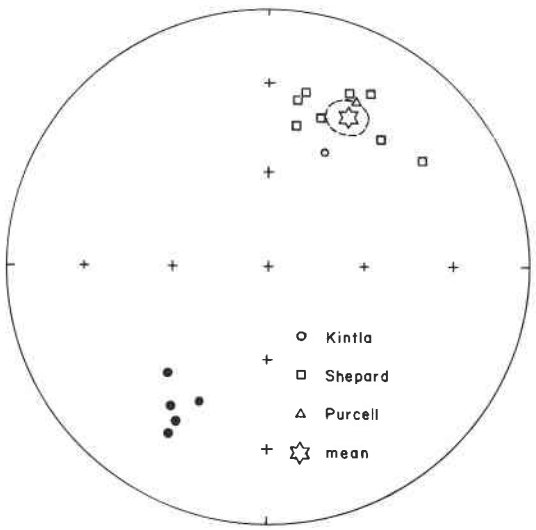


FIG. 1. Mean site directions after thermal cleaning and correction for geological dip. Equal-area stereographic projection; solid (open) symbols represent north-seeking directions on the lower (upper) hemisphere. The cone of 95% confidence around the mean is indicated. See Tables 1 and 2.

always underestimates the circle of confidence (Irving 1964). The presence of hematite suggests that thermal demagnetization will remove secondary components of magnetization more effectively than magnetic treatment. This is reflected in the somewhat tighter grouping obtained by the former procedure (cf. entries 5 and 6, Table 2). The thermal result is therefore used henceforth. It diverges from Black's (1963) result by about 5°, which is not significant since Black's data are highly scattered ($N = 22$ samples, $k = 4$, $\alpha_{95} = 18.5°$).

(b) Shepard Formation

The Shepard Formation overlies the Purcell Lava with a sharp contact. It consists mostly of light-colored beds of dolomite and sandstone, but some horizons are red hematitic sandstone. Sampling was concentrated in these red strata.

Two specimens from each site were selected for detailed pilot demagnetization studies; one to be studied by thermal methods and one by magnetic methods. In general there appear to be no decisive differences between the two methods, although in all cases thermal treatment is more effective in reducing the intensity of magnetization. This is typical of results from red sediments which usually contain magnetically hard hematite. With this in mind we have subjected remaining specimens to thermal treatment only, but it is perhaps worth pointing out that pilot studies indicate that essentially the same directional information results from a.f. treatment (Table 2, entries 3 and 4).

Thermal demagnetization curves exhibit a variety of behaviors, suggesting a range from discrete to distributed blocking temperature spectra. The intensities after 600°C treatment range from 2% to 103% of the NRM values but in all cases the directions remain stable indicating that hematite is probably the dominant

magnetic mineral. During heating, movements of the remanence vectors were generally small. Angular shifts of 25° and less occur between NRM and 500°C, thereafter good "end-points" exist up to 660°C (in two cases, GAB and GAC, to 690°C) with further angular shifts never exceeding 10°. On the basis of these pilot results, thermal treatments at 500 and 600°C were selected for "cleaning" a total of two specimens per hand block. Two treatments were used as an extra check on the directional stability of the remanence. The results were averaged to obtain sample means for each treatment, which were then combined to yield site means in the normal hierarchical manner. There is no essential difference between the 500 and 600°C results, individual site means never differing by more than 10°. The results for the higher temperature are quoted as they are always slightly further away from the present field direction at the sampling locality, which suggests that recently acquired secondary components of magnetization have been somewhat more effectively removed.

Sites from the Shepard Formation span 120 stratigraphic meters and all are of reversed polarity. The statistical results are summarized in Table 1, and the site means are illustrated in Fig. 1.

(c) Kintla Formation

In the region under discussion the Kintla Formation has been informally divided into four members (A, B, C, and D) by Hage (1943), member A being the lowermost. Member A consists predominantly of red and purplish-red siltstone, whilst member C consists mostly of red and purplish-red quartz sandstone. These members were previously sampled by Black (1963) and consequently we have concentrated our sampling on member B (Fig. 2) (member D does not outcrop in the area sampled). This member is predominantly green and greenish grey micaceous argillite, but occasional maroon horizons occur, and most sites are located in such horizons. However, two of the B-member sites are located in green horizons and do not respond simply to routine laboratory techniques; they are discussed separately below. For the other six sites pilot specimens were treated in the same way as for the Shepard Formation. Again thermal and a.f. treatments yield virtually identical directional data (Table 2, entries 1 and 2). Before treatment NRM intensities range from 1×10^{-5} to 1×10^{-4} emu/cm^3. Thermal demagnetization curves are generally "square-shouldered" with 45% to 99% of the NRM remaining at 600°C, again indicating the importance of hematite. Between NRM and 500°C no angular shift greater than 12° takes place, and between 500 and 660°C no shift greater than 10° occurs. In fact little directional change occurs up to 690°C, the maximum temperature used. Where material was available (some samples were small and friable and tended to break apart during laboratory preparation), a total of two specimens per sample were heated to 500 and 600°C. Site means for the two temperatures never differ by more than 9°, and therefore only one set of results is reported (Table 1). The site mean directions are illustrated in Fig. 1.

The two sites exhibiting anomalous behavior are sites GBD and GBF. Both sites are located in green horizons and NRM intensities are weak, ranging from 3×10^{-7} to 2×10^{-6} emu/cm^3. Thermal demagnetization curves indicate a wide range of blocking temperatures decreasing more or less linearly to 550°C, after which directions become scattered and intensities barely measureable. These two green sites therefore yield no clear evidence for the presence of hematite.

Some specimens from site GBD show reasonable "end-points" after magnetic cleaning, but others are magnetically unstable. Under thermal demagnetization two specimens from each sample yielded very good within-sample agreement at 400°C, but the between-sample agreement is poor (Table 1).

Material from site GBF behaves in a variety of ways during cleaning. Some specimens show clear "end-points" but others are very unstable. Specimens from sample GBF02 agree closely with each other in fields up to 100 Oe and temperatures up to 450°C (*e.g.*, $k = 142$ at 400°C), but specimens from sample GBF01 possess widely different remanence directions.

The results obtained from the only samples from these two sites exhibiting good within-sample agreement (*i.e.* GBD01, GBD02 and GBF02) are included in Table 1. In view of the scattered results and variable response to cleaning, results from sites GBD and GBF are rejected as reliable indicators of the ancient geomagnetic field. It is worth pointing out, however, that this exclusion does not affect the final conclusions in any important way as inclusion

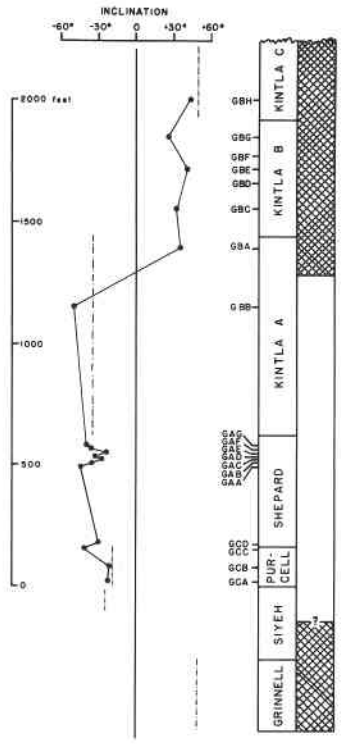


FIG. 2. Stratigraphic distribution of sampling sites and site mean inclinations. The mean inclinations found by Black (1963) are shown as dash–dot lines. The actual stratigraphic horizons sampled were not recorded by Black. Note that his Kintla B result was derived from an intruded sill (see text). Vertical scale applies only where indicated.

of the "anomalous" data shifts the overall result by only 4°.

The Kintla sampling sites span 260 stratigraphic metres and all but the lowest (site GBB) possess normal polarity. The mean direction for the six reliable Kintla sites (entry 1, Table 2) does not differ significantly from the results obtained by Black (1963) from the Kintla A and C members. His combined result, obtained by inverting the polarity of the reversed samples, is $D = 223°$, $I = 44°$, $k = 15$, $\alpha_{95} = 6°$, $N = 38$ samples.

Discussion

The stratigraphic arrangement of the sites is shown in Fig. 2, along with the inclination of the site mean remanence directions. In the lower half of the section the rocks possess normal polarity, but this switches to reversed near the top of member A of the Kintla Formation. The reversal behavior of the geomagnetic field in these remote times is thus beginning to emerge, although much more detailed stratigraphic work is required to cover the whole of the Belt – Purcell Supergroup. Included in Fig. 2 is a summary of Black's (1963) data from various Belt – Purcell Supergroup formations in Alberta and British Columbia. It must be remembered that Black does not report the stratigraphic relationships between samples *within* given formations. The magnetic results indicate normal polarity down to the upper Kintla A Formation[1], prior to which a reversed polarity period occurred. The base of this latter period is not yet fixed, although it must lie somewhere in the Siyeh (Helena) Formation as Black reports reversed polarity from the uppermost 30 m of the Siyeh Formation whereas the underlying Grinnell (Spokane) Formation is normal (Black 1963; Collinson and Runcorn 1960). Detailed work currently being prepared for publication indicates that normal polarity exists throughout the Grinnell Formation (except for a single reversed site near the bottom of the Grinnell) and into the lowermost Siyeh Formation.

In any paleomagnetic study it is desirable to seek evidence for the age of the remanence observed, and this is particularly true in the case of Precambrian sediments. The evidence for deuteric hematite in the Purcell Lava suggests that the remanence in that formation is primary. Also, the anti-parallel relationship of the normal and reversed directions (compare entries 7 and 8 of Table 2) and the divergence from the present field vector at the sampling site (present field is $D = 20°$, $I = +73°$, cf. entry 9, Table 2) provide evidence against substantial secondary magnetizations of recent origin masking a primary remanence in the Shepard and Kintla Formations. Furthermore, the pattern of polarity reversals revealed by these and other Belt formations provides evidence for the primary nature of the magnetization involved. It is unlikely that such a simple, coherent pattern could be produced by any magnetization process which operated at a time much later than that of original deposition.

Hunt (1962) reports an age of 1075 (±6%) m.y. on a hornfels produced by the Purcell Lava. Obradovich and Peterman (1968) have obtained

[1]Black obtained a reversed direction from a gabbroic sill in the Kintla B Formation, and this implies a reversed period sometime after Kintla C time.

a single isochron age of 1097 ± 20 m.y. for samples representing a considerable stratigraphic interval, from below the Siyeh (Helena) Formation up to the McNamara Formation, which lies above the Kintla (Mount Shields) Formation. Using the experimental uncertainty as a measure of the time interval represented, they claim that "40 m.y. could reasonably be inferred for the time span of deposition of this sequence." (Obradovitch and Peterman 1968, p. 744). Harrison (1972) suggests that the known sequence of geological events makes it unlikely that this single isochron correctly represents the ages of the entire sequence. Instead he argues that the 1100 m.y. age is reasonably well documented for the lower part of the Missoula Group with part of which the Purcell Lava, Shepard and Kintla Formations are correlated, but it is likely that the lower formations will eventually prove to be older. In any case the implied frequency of polarity reversals is at least an order of magnitude lower than that observed over the last few million years of geological time (Cox 1969), since there appear to be only two reversals in the stratigraphic sequence from Grinnell to Kintla C Formation, which apparently represents at least 40 m.y. of time and probably as much as 200 m.y. (see Harrison 1972, Fig. 15). These infrequent polarity changes have obvious relevance to the history of reversals and consequent implications for geomagnetic dynamo behavior (Cox 1968), and they also possess considerable potential for regional stratigraphic correlation.

Quite apart from the interval of time occupied by these reversals, it is essential for the purposes of establishing a reliable polar wandering curve to consider their absolute age. Harrison's review suggests an age of about 1100 m.y. but an older age cannot be ruled out. Reid *et al.* (1973) argue that their geochronological data imply an age of about 1500 m.y. for the formations up to the Siyeh (Helena), whereas Obradovich and Peterman (1968) and Harrison (1972) prefer an age of about 1300 m.y. for these strata. Ryan and Blenkinsop (1971) obtain an age of 1260 ± 50 m.y. for the Hellroaring Creek stock of British Columbia; Obradovich and Peterman (1973) recalculate this age to obtain 1335 m.y. This granodioritic intrusion cuts the Aldridge Formation which lies close to the bottom of the Belt – Purcell Supergroup. Also intruding the Aldridge Formation are a series of basic sills referred to as the Moyie intrusions; these are also cut by the Hellroaring Creek stock and hence pre-date it. Now, Moyie intrusions outside the area of exposure of the Hellroaring Creek stock intrude strata as high as the Siyeh Formation. Ryan and Blenkinsop consequently infer two ages of Moyie intrusions, but this is not firmly established. If only one intrusive phase occurred then the formations up to and including the Siyeh must be older than about 1300 m.y. which conflicts with the data of Obradovich and Peterman (1968).

To summarize, this brief discussion of the available radiometric data indicates that the formations studied paleomagnetically in this paper are regarded by some authors as being about 1100 m.y. old, but an age some 200 m.y. greater is not impossible. This point is important for the discussions below where the pole position obtained is examined in the context of the overall polar wandering curve relative to North America.

Three stratigraphically distinct formations have been studied. The mean directions of remanence obtained from these formations are given separately in Table 2 (entries 1, 3 and 5). Taken in pairs these formation mean directions are separated by 7° (Shepard-Kintla), 12° (Shepard-Purcell), and 16° (Kintla-Purcell), and an *F*-ratio test following Watson (1956) indicates that even the 7° separation is marginally significant at the 95% level despite the considerable overlap of the appropriate α_{95} circles. Furthermore, there is a trend from shallow inclination in the lowest formation (Purcell Lava) to steep inclination in the highest formation (Kintla). It may be therefore that the results record a small, but real, change in pole position with time. However, it is stressed that these differences are small and their possible geophysical implications should not be over-emphasized at this stage. With this in mind, we here take the conservative approach of combining the fifteen site means (Table 2, entry 10) to obtain a paleomagnetic pole position for the time interval covered.

Before correction for the present attitude of the beds the 15 site means have a precision parameter of 33.1 (entry 11, Table 2), compared to 39.4 after correction. This improvement in the dispersion is not large enough to be significant at the 95% level and thus the fold test (Graham 1949) is indecisive in establishing the relative

ages of the remanence and the folding. This situation arises from the fact that 13 of the sites (from the ridges southwest of Mt. Gladstone) comprise a conformable uniformly-dipping sedimentary sequence (Table 1). Thus these sites, representing most of the data, dominate the statistics of the whole population. A more instructive approach is to consider the agreement between paleomagnetic poles obtained from upper Belt formations collected from several localities throughout the Belt basin and presumably, therefore, possessing widely varying attitudes (see Fig. 3).

A second point concerning the local and regional geological structure merits attention: namely, possible rotation (about a vertical axis) associated with movement during the Laramide orogeny of the Lewis thrust plate. Norris and Black (1962) have discussed this problem in detail and conclude that there is no significant differential rotation between the Flathead and Clark Ranges, both of which lie north of the 49th parallel. Agreement between upper Belt poles from widely spaced localities (Table 3) supports their conclusion. Furthermore, lower Belt strata yield paleomagnetic poles concordant with those obtained from time-equivalent rocks throughout North America (see below). These points suggest that major tectonic effects have been properly allowed for; small effects ($\lesssim 10°$) may, of course, be present, but these would scarcely be detectable by paleomagnetic methods.

The overall mean direction ($D = 208.3°$, $I = +34.7°$) yields a paleomagnetic pole at 142.2°W, 17.1°S ($dp = 4.1°$, $dm = 7.1°$), on the assumption of a geocentric dipole field. This implies that the sampling area (now at 114°W, 49°N) was situated at latitude 19.1° at the time the magnetization was acquired, which is consistent with the abundant occurrence of mudcracks and halite casts, indicative of a warm, desiccating climate (Price 1964).

The paleomagnetic pole deduced from the present study is shown in Fig. 3 with other relevant results from North America including earlier Belt results. There is a clear grouping of poles in the equatorial mid-Pacific which has been recognized by many workers and is referred to by Spall (1971) as the "Elsonian trend". Averaging these poles in the manner shown in Table 3 one obtains a mean pole at 146°W, 6°S ($A_{95} = 6°$), and for the purpose of discussion



FIG. 3. Paleomagnetic pole for the Missoula Group (triangle), compared with other relevant North American poles. Ages are in millions of years. The data from which the Elsonian pole is derived are given in Table 3. The five poles defining the "Logan Loop" and the ages assigned to them are from Robertson and Fahrig (1971). Equal area stereographic projection.

we refer to this as the "Elsonian pole". It is important to note that this group of data represents results from several geological provinces of the Canadian Shield, plus results from Missouri, Colorado and the Belt basin. This clustering consequently provides evidence that many of the individual structural units of the present North American continent were probably a single integral entity at this early time. Spall associates an age range of 1300 to 1475 m.y. with this "trend"; consequently, we have included in the Elsonian pole only *lower* Belt – Purcell Supergroup results which can be fairly certainly associated with this interval. However, the results presented here are derived from *upper* Belt – Purcell Supergroup formations which may be considerably younger. Despite this apparent age difference the geomagnetic pole lies close to the Elsonian pole, as does the well-defined average of previously published upper Belt results (Table 3 and Fig. 3).

About 30° West of the Elsonian pole there is a second distinct group of paleomagnetic poles, first recognized by Fahrig and Jones (1969) and referred to by Spall (1971) as the "Mackenzie trend". Fahrig and Jones conclude that the age of this group is 1200 m.y. Consequently, most

TABLE 3. Summary of "Elsonian" paleomagnetic data

Unit	Symbol	Pole position	95% conf. oval/circle	Age	Comments
Croker Island Complex, Ontario	CI	143W 5N	7, 11	1475 ± 50	010129
Front Range Rocks, Colorado–Wyoming	FR	151W 8S	4, 6	1410 ± 30	010172
Michikamau Anorthosite, Labrador	MA	145W 1S	5, 10	1400 ± 80	010205
Sibley Group, Ontario	SG	144W 21S	7, 13	1370	Robertson 1973
St. Francois Mt. Rocks, Missouri	SF	143W 1S	8, 12	1300–1400	010241
Lower Belt Supergroup ($N = 4$)	LB	148W 8S	16	1325 ± 15?	notes 1, 3
Elsonian Pole ($N = 6$)		146W 6S	8	1300–1475	mean of 6 entries above
Upper Belt Supergroup ($N = 6$)	UB	152W 16S	9	1097 ± 20?	notes 2, 3

NOTES: Symbol, refers to designation in Fig. 3.
95% conf. oval/circle, refers to dp, dm or A_{95} (Irving 1964).
Six-digit figures are those assigned by Hicken *et al.* (1972).
1. Lower Belt Supergroup pole obtained by meaning results from Grinnell (mean of 010025 and 010070), Appekunny (010026), Spokane (010023), and Waterton (010071) formations ($N = 4$).
2. Upper Belt Supergroup pole obtained by meaning results from Miller Peak Fm. (010022, McNamara Fm. (010021), Missoula Group (010224), Purcell Lava (010068 and pole derived from entry 5, Table 2), Shepard Fm. (pole derived from entry 3, Table 2), and Kintla Fm. (010067, 010065, and pole derived from entry 1, Table 2) ($N = 6$).
3. The ages quoted are from Obradovich and Peterman (1968); they may be 100 to 200 m.y. too young (see text).

authors (*e.g.*, Robertson and Fahrig 1971; Irving and Park 1972) draw the path of apparent polar motion relative to North America *from* the Elsonian pole *to* the Mackenzie pole. Current ideas then require a large northward loop in the polar path, the "Great Logan Loop" of Robertson and Fahrig (1971). These points are summarized in Fig. 3.

It is clear that a conflict exists between the data used to define the "Logan Loop" and that obtained from the Belt basin, since Belt Supergroup poles apparently as young as 1100 m.y. lie very close to the Elsonian pole as defined above whereas the Logan Loop interpretation requires a much more northerly position for the pole at this time. There are at least four possible explanations:

(a) The lower part of the Missoula Group pre-dates the Mackenzie pole, which is regarded by Fahring and Jones as about 1200 m.y. old. In terms of the paleomagnetic results above, this interpretation is the most attractive and in view of the current debate concerning Belt geochronology it cannot be dismissed. It requires a revision of Harrison's (1972) suggested chronology to make the appropriate strata 100 to 200 m.y. older; this is consistent with the views of at least some authors.

(b) At the appropriate time, or times, the geomagnetic field was not dipolar. For the Elsonian data ($\gtrsim$ 1300 m.y.) and the Mackenzie data ($\sim$ 1200 m.y.) this is unlikely because the concordant data leading to these poles comes from widely spaced regions of the North American continent. But the possibility remains that the field was not dipolar for an interval of time around 1100 m.y. ago. As Irving (1964 p. 133) points out this so-called non-dipole hypothesis is non-scientific and serves no purpose until the simple refutable hypotheses have been tested.

(c) The pole moved from the Elsonian position to the Mackenzie and then back again before executing a rapid excursion northwards through the Abitibi pole and around the "Logan Loop". This possibility embraces minor age inconsistencies since the Abitibi dike swarm is thought to be 1150 m.y. old (see Fig. 3). However, the age of these dikes is not established beyond doubt and it is possible that they post-date the youngest Belt sediments.

(d) Perhaps the most interesting possibility, geologically speaking, it that of relative motion between the "sub-plates" currently comprising the North American continent. In this view the Belt region retained a fixed position relative to the pole from about 1300 m.y. to 1100 m.y., whereas the other provinces yielding "Elsonian" poles moved in an appropriate manner to describe the "Logan Loop". Further work is needed to assess this possibility, but it should be pointed out that there is no geological evidence in the basement underlying the Alberta Basin which supports post-Belt activity (Burwash *et al.* 1964). Other workers have made similar suggestions concerning various provinces of the Canadian Shield (Irving *et al.* 1972; Gibb and Walcott 1971). Such ideas represent the natural

consequences of plate tectonic processes operating in Precambrian times.

Some of these explanations imply considerable complexity in the paleomagnetic record. It will therefore be of some interest to see if further paleomagnetic work can distinguish between the various possibilities, and perhaps determine the actual sequence of events which took place.

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